

1. Administrative & Study Identification

Full Study Title: A Study to Investigate Efficacy, Safety, and Tolerability of Remibrutinib Compared With Placebo in Adults With CINDU Inadequately Controlled by H1-antihistamines **Official Regulatory Title:** A 52-week Multi-center, Randomized, Double-blind, Placebo Controlled, Basket Study With an Open-label Extension to Investigate the Efficacy, Safety, and Tolerability of Remibrutinib (LOU064) in Chronic Inducible Urticaria (CINDU) in Adults Inadequately Controlled by H1-antihistamines **Short Title/Acronym:** RemIND trial **Protocol Number:** CLOU064M12301 **NCT Number:** NCT05976243 **Trial Status:** Active, Not Recruiting **Sponsor/Company:** Novartis **Investigational Product:** Remibrutinib (LOU064) **Phase of Development:** Phase 3 **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy, safety, and tolerability of remibrutinib vs. placebo in adults suffering from Chronic Inducible Urticaria (CIndU) inadequately controlled by H1-antihistamines, specifically assessing the proportion of complete responders at Week 12. **Secondary Objectives:** To evaluate long-term efficacy, safety, and tolerability during the 52-week core period and up to 3 years in the optional Open-label Extension (OLE).

2.2 Background & Rationale CIndU is a chronic skin condition affecting approximately 0.5% of the population, characterized by hives and/or swelling triggered by identifiable external stimuli like pressure, cold, heat, or friction. Patients often face a significant burden, cycling through antihistamines without adequate relief. Remibrutinib is a highly selective, oral Bruton’s tyrosine kinase (BTK) inhibitor designed to block the BTK pathway involved in the release of histamine—a central driver of wheals and swelling—thereby providing targeted relief where no targeted therapies have previously been approved for this condition.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional, Confirmatory **Control Type:** Placebo-controlled **Blinding:** Double-blind (during the 24-week core treatment period), transitioning to Open-label (28 weeks) **Randomization:** Randomized, Parallel group, Basket study (6 arms in the Core period)

3.2 Planned Enrollment & Duration Targeted Enrollment: 364 participants **Number of Sites:** Global, Multicenter **Estimated Study Start:** 07-Dec-2023 **Estimated Primary Completion:** 22-May-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria for Core Period 1. Male and female participants ≥ 18 years of age at the time of signing of the ICFs. 2. Confirmed CIndU diagnosis for symptomatic dermographism, cold urticaria, or cholinergic urticaria for ≥ 4 months and inadequate control with H1-AHs at local label approved doses at the time of randomization. 3. The following responses to provocation tests at randomization: * **Symptomatic Dermographism:** Total Fric Score ≥ 3 (FricTest® 4.0) and itch NRS score ≥ 5 . * **Cold Urticaria:** Critical Threshold Temperature (CTT) $\geq 15^{\circ}\text{C}$ (TempTest® 4.0) and itch NRS score ≥ 5 . * **Cholinergic Urticaria:** Physician Global Assessment of severity of hives ≥ 2 (Pulse-controlled ergometry test) and itch NRS score ≥ 5 . 4. **Cold Urticaria specific:** Positive ice-cube test resulting in hives at the provocation site at Screening. 5. **Cholinergic Urticaria specific:** Participants must show sweating in performing the pulse-controlled ergometry test on day of randomization (participants with anhidrosis must not be included).

4.2 Inclusion Criteria for Open-Label Extension (OLE) 1. Participants who have completed the Core period up to Week 52 and are willing to enter the OLE period.

4.3 Exclusion Criteria for Core Period 1. Previous use of remibrutinib or other BTK inhibitors. 2. Concomitant Chronic Spontaneous Urticaria (CSU) at screening (resolved CSU is allowed). 3. Familial form of the target CIndU (e.g., familial cold autoinflammatory syndrome). 4. A more defined other form of inducible urticaria than the target CIndU. 5. Diseases (other than CIndU) with urticaria/angioedema symptoms (e.g., urticarial vasculitis, erythema multiforme, cutaneous mastocytosis, hereditary/acquired angioedema). 6. Any other skin disease associated with chronic itching that might influence study evaluations (e.g., atopic dermatitis, bullous pemphigoid), or skin diseases associated with only wheals and no itch (e.g., asymptomatic dermographism). *(Note: There are no exclusion criteria for the OLE)*

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Remibrutinib administered orally versus matching placebo. **Duration of Treatment:** * **Core Period:** Up to 60 weeks total (Up to 4 weeks screening, 24 weeks double-blind treatment, 28 weeks open-label remibrutinib, and 4 weeks of treatment-free follow-up). * **Open-Label Extension (OLE):** Approximately 3 years. Participants who

continue to experience symptoms transition to the *OLE Treatment Period* (remibrutinib). Participants who do not experience symptoms transition to the *OLE Observation Period* (no remibrutinib) but can switch to treatment if symptoms develop. **Pharmacology Note:** ATC Code L01EL01 classification logic (BTK inhibition analog, e.g., ibrutinib).

5.2 Concomitant & Prohibited Therapy Permitted: Standard H1-antihistamines as background therapy. **Prohibited:** Other BTK inhibitors or investigational targeted biologics.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	Up to 4 weeks	Informed Consent, Medical History, CIndU subtype provocation tests, Ice-cube test (Cold Urticaria)
2	Day 0	Randomization, First Dose, Provocation Tests (ergometry sweating check)
3	Week 12	Primary Endpoint Assessment (Provocation Testing)
4	Week 52	Final Efficacy/Safety Assessment for Core Phase, transition to OLE
5	Week 56	4-week treatment-free follow-up (if not entering OLE)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints (Assessed at Week 12)

- Symptomatic Dermographism:** Proportion of participants with complete response in Total Fric Score (a scale from 0-4 where a positive response with all four pins is TFS = 4, and only the largest pin is TFS = 1).
- Cold Urticaria:** Proportion of participants with complete response in Critical Temperature Threshold (CTT). The TempTest determines the highest temperature that induces symptoms.
- Cholinergic Urticaria:** Proportion of participants with itch numerical rating scale (NRS) = 0. Patients rate worst itch severity in the past 24 hours on an 11-point scale from 0 ("no itch") to 10 ("worst itch imaginable").

7.2 Secondary Endpoints

- Change from baseline in Total Fric Score (Symptomatic Dermographism).
- Change from baseline in Critical Temperature Threshold following TempTest (Cold Urticaria).
- Change from baseline in Itch Numerical Rating Scale (Cholinergic Urticaria).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring via patient electronic diaries, routine onsite evaluations, and HCP interviews. **Serious Adverse Events (SAEs):** Standard protocol reporting timeline. **Data Monitoring Committee (DMC):** Internal/

External independent safety review committee per Novartis protocols.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy outcome measurements. Safety Set for all participants receiving at least one dose of the investigational product. **Sample Size Justification:** Targeted enrollment of $N = 364$ participants across the CIndU subtype cohorts to adequately power the primary endpoint comparisons at Week 12. **Handling of Missing Data:** Standard multiple imputation techniques, with non-responder imputation for missing binary outcomes.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: Conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Remote and onsite monitoring to ensure integrity of standardized provocation testing. **Audit Trail:** Secure eCRF systems, provocation device calibration logs, and standardized data clarification protocols.

11. Study Identifiers & Governance

Primary ID: CLOU064M12301 **Posting ID:** 526688267967951255 **Registry Number:** NCT05976243 **Sponsor/Lead Organization:** Novartis **Study Title:** RemIND trial **Official Regulatory Title:** A 52-week Multi-center, Randomized, Double-blind, Placebo Controlled, Basket Study With an Open-label Extension to Investigate the Efficacy, Safety, and Tolerability of Remibrutinib (LOU064) in Chronic Inducible Urticaria (CINDU) in Adults Inadequately Controlled by H1-antihistamines

12. Clinical Framework (CIndU Specific)

Disease Area: Chronic Inducible Urticaria (MeSH: D000094482, SNOMED: 712211002) **Diagnosis Threshold:** Inadequate control with H1-AHs and positive subtype-specific provocation tests. **Medical Methodology:** Bruton's Tyrosine Kinase (BTK) inhibition. **Optimization Target:** Complete symptomatic relief and absence of hives upon provocation.

13. Study Procedures & Methodology

Management Pathway: Targeted CIndU symptom management and strict provocation threshold testing. **Education Protocol (HCP):**

Specialized training for standardized provocation tests (FricTest® 4.0, TempTest® 4.0, Pulse-controlled ergometry).
Education Protocol (Patient): Self-learning on identifying triggers and maintaining accurate electronic symptom diaries (0-10 itch NRS, 24-hour worst recall). **Quality Audit Mechanism:** Device calibration tracking for provocation tests & continuous patient symptom diary compliance.

14. Observation & Follow-Up Schedule

Total Duration: Up to 60 weeks core + up to 3 years Open-label Extension (incorporating treatment and observation periods).
Onsite Visit Cadence: Screening, Baseline, Week 12, Week 24, Week 52, and specific OLE interval visits. **Checklist Review Interval:** Daily symptom tracking via e-diaries evaluating the worst level of itching in the past 24 hours.

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				